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ELECTRO-OPTICAL DEVICE AND ELECTRONIC APPARATUS

Abstract

An electro-optical device includes a substrate, a pixel electrode, a light-emitting layer in contact with the pixel electrode in plan view at an opening region defined by an opening end, a common electrode that sandwiches the light-emitting layer with the pixel electrode, a conductive partition wall being in contact with the common electrode and surrounds the pixel electrode, the light-emitting layer, and the common electrode in plan view, and an upper portion provided on the upper surface of the partition wall to protrude from the partition wall in cross-sectional view. When a distance of a Z-direction component of a straight line coupling a tip of the upper portion to the opening end is defined as α and a distance of a component of the straight line along a substrate surface is defined as β , there is a relationship of $\alpha > \beta$.

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Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-026354, filed Feb. 26, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an electro-optical device and an electronic apparatus.

2. Related Art

[0003] An electro-optical device using, for example, an OLED as a display element is known. The OLED is an abbreviation for Organic Light Emitting Diode. Such a light-emitting element has a configuration in which a light-emitting layer is sandwiched between a pixel electrode and a common electrode. Since the common electrode is common to all pixel portions, it is necessary to curb voltage unevenness due to resistance components. Therefore, a technology for surrounding the pixel electrode, the light-emitting layer, and the common electrode in plan view with a conductive partition wall and applying a voltage to the common electrode via the conductive partition wall is known.

CITATION LIST

Patent Literature

[0004] [PTL 1] JP-A-2023-100414

[0005] However, when an interval between pixel portions is narrowed to a few μm , there is a problem that light emitted from the light-emitting layer is shielded by an upper portion provided on an upper surface of the partition wall, or an aperture ratio in the pixel portion is reduced.

SUMMARY

[0006] An advantage of some aspects of the disclosure provides an electro-optical device including a substrate, a first electrode, a second electrode provided between the substrate and the first electrode, a pixel separation layer having an insulative property, and covering the periphery of the second electrode and opening in an opening region overlapping the second electrode in plan view, a light-emitting layer provided between the first electrode and the second electrode and being in contact with the second electrode in the opening, a partition wall having a light-shielding property against light emitted by the light-emitting layer and surrounding the first electrode, the light-emitting layer, and the second electrode in plan view, and an upper portion provided so as to protrude from the partition wall in cross-sectional view on an upper surface of the partition wall, and having a light-shielding property against light emitted by the light-emitting layer, wherein, when a distance of a normal component to the substrate of a shortest straight line coupling a tip of the upper portion to an opening end of the opening region in cross-sectional view is defined as α , and a distance of a component of the straight line along a surface of the substrate is defined as β , there is a relationship of $\alpha > \beta$.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a perspective view illustrating a configuration of an electro-optical device according to a first embodiment.

[0008] FIG. 2 is a diagram illustrating an electrical configuration of the electro-optical device.

[0009] FIG. **3** is a diagram illustrating a configuration of a pixel circuit in the electro-optical device.

[0010] FIG. **4** is a diagram illustrating an operation of the electro-optical device.

[0011] FIG. **5** is a plan view illustrating main portions of a pixel portion in the electro-optical device.

[0012] FIG. **6** is a partial cross-sectional view schematically illustrating the electro-optical device.

[0013] FIG. **7** is a partial cross-sectional view illustrating the electro-optical device.

[0014] FIG. **8** is a partial cross-sectional view illustrating the electro-optical device.

[0015] FIG. **9** is a partial enlarged cross-sectional view illustrating the electro-optical device.

[0016] FIG. **10** is a diagram illustrating a relationship between an observation angle and a relative luminance in the electro-optical device.

[0017] FIG. **11** is a partially enlarged cross-sectional view illustrating the electro-optical device.

[0018] FIG. **12** is a partially enlarged cross-sectional view illustrating an electro-optical device according to a first embodiment.

[0019] FIG. **13** is a partially enlarged cross-sectional view illustrating an electro-optical device according to a second embodiment.

[0020] FIG. **14** is a partially enlarged cross-sectional view illustrating an electro-optical device according to a third embodiment.

[0021] FIG. **15** is a partially enlarged cross-sectional view illustrating an electro-optical device according to a comparative example.

[0022] FIG. **16** is a partially enlarged cross-sectional view illustrating an electro-optical device according to a first application example.

[0023] FIG. **17** is a partially enlarged cross-sectional view illustrating an electro-optical device according to a second application example.

[0024] FIG. **18** is a perspective view illustrating a head-mounted display using the electro-optical device.

[0025] FIG. **19** is a diagram illustrating an optical configuration of the head-mounted display.

DESCRIPTION OF EMBODIMENTS

[0026] Hereinafter, an electro-optical device according to an embodiment will be described with reference to the accompanying drawings. In each drawing, dimensions and scales of each portion are appropriately different from actual ones. Further, since embodiments to be described below are preferred specific examples, various technically preferable limitations are applied, but the scope of the present disclosure is not limited to these embodiments unless it is otherwise stated in the following description that the present disclosure is limited.

[0027] FIG. **1** is a perspective view illustrating an electro-optical device **10** according to a first embodiment, and FIG. **2** is a block diagram illustrating an electrical configuration of the electro-optical device **10**.

[0028] The electro-optical device **10** is a micro display panel that displays color images, for example, in a head-mounted display. The electro-optical device **10** includes a plurality of pixel portions or a driving circuit that drives the pixel portions. The pixel portions and the driving circuit are integrated on a semiconductor substrate. The semiconductor substrate is typically a silicon substrate, but may be another semiconductor substrate.

[0029] The electro-optical device **10** is accommodated in a frame-shaped case **192** that opens in the display region **100**. One end of an FPC board **194** is coupled to the electro-optical device **10**. FPC is an abbreviation for Flexible Printed Circuit. The other end of the FPC board **194** is provided with a number of terminals **196** for coupling to a host device ((not illustrated)). When a plurality of terminals **196** are coupled to the host device, video data, a synchronization signal, and the like are supplied from the host device to the electro-optical device **10** via the FPC board **194**.

[0030] As illustrated in FIG. **2**, the electro-optical device **10** is broadly divided into a control circuit **30**, a data signal output circuit **50**, a display region **100**, and a scanning line driving circuit **120**.

[0031] In the display region **100**, m rows of scanning lines **12** are provided along an X direction, and $(3n)$ columns of data lines **14** are provided along a Y direction to be electrically insulated from each of the scanning lines **12**. m is an integer equal to or greater than 2, and n is an integer equal to or greater than 2.

[0032] To generalize and explain the scanning lines **12**, an integer i of 1 or more and m or less is used. To distinguish rows of the scanning lines **12**, the rows may be referred to as first, second, third, . . . , i -th, . . . , $(m-1)$ -th, m -th row from the top in the figure.

[0033] Similarly, an integer j of 1 or more and n or less is used to generalize and explain the data lines **14**. To distinguish columns of the data lines **14**, the columns may be referred to as first, second, third, . . . , $(3j-2)$ -th, $(3j-1)$ -th, $(3j)$ -th, . . . , $(3n-2)$ -th, $(3n-1)$ -th, and $(3n)$ -th columns from the left in the figure.

[0034] In the display region **100**, a pixel portion **110R** that emits light in a red wavelength region, a pixel portion **110G** that emits light in a green wavelength region, and a pixel portion **110B** that emits light in a blue wavelength region are provided in the following manner to correspond to intersections of the m -th row of scanning line **12** and the $(3n)$ -th column of data line **14**.

[0035] The pixel portion **110R** is provided to correspond to an intersection of the scanning line **12** of each row and the $(3j-2)$ -th column of data line **14**. The pixel portion **110G** is provided to correspond to an intersection of the scanning line **12** of each row and the $(3j-1)$ -th column of data line **14**. The pixel portion **110B** is arranged to correspond to an intersection of the scanning line **12** of each row and the $(3j)$ -th column of data line **14**.

[0036] That is, in the display region **100**, pixel portions **110R**, **110G**, and **110B** are arranged in this order along the X direction. A single color is expressed by additive color mixing of three pixel portions **110R**, **110G**, and **110B** adjacent in the X direction. Thus, the electro-optical device **10** displays an image in which color pixels are arranged in m rows and n columns.

[0037] Strictly speaking, the pixel portions **110R**, **110G**, and **110B** should be called sub-pixel portions, but are referred to as pixel portions for convenience of description. Further, when the pixel portions **110R**, **110G**, and **110B** are generally described without specifying the color, the pixel portions **110R**, **110G**, and **110B** are denoted by a reference number **110**.

[0038] The control circuit **30** controls each part based on the video data Vid or the synchronization signal Sync supplied from a host device (not illustrated). Specifically, the control circuit **30** generates various control signals to control the respective parts.

[0039] The video data Vid designates the gradation level of the pixel in the image to be displayed, for example, in 8 bits. The synchronization signal Sync includes a vertical synchronization signal for giving an instruction for starting vertical scanning of the video data Vid, a horizontal synchronization signal for giving an instruction for starting horizontal scanning, and a dot clock signal that indicates a timing of one pixel of the video data.

[0040] The luminance characteristics at the gradation level indicated by the video data Vid supplied from the host device do not necessarily match the luminance characteristics of the OLED included in the pixel portion **110**. Thus, to make the OLED emit light at a luminance corresponding to the gradation level indicated by the video data Vid, the control circuit **30** up-converts 8 bits of the video data Vid into, for example, 10 bits and outputs it as video data Vdata. Therefore, the 10-bit video data Vdata becomes data corresponding to R, G, and B gradation levels designated by the video data Vid.

[0041] For the upconversion, a lookup table in which a correspondence relationship between 8 bits of the video data Vid that is an input and 10 bits of the video data Vdata that is an output is prestored is used.

[0042] The scanning line driving circuit **120** is a circuit for driving the pixel portions **110** arranged in m rows and $3n$ columns one by one under the control of the control circuit **30**. For example, the scanning line driving circuit **120** supplies scanning signals $/Gwr(1)$, $/Gwr(2)$, . . . , $/Gwr(m-1)$, $/Gwr(m)$ to the scanning lines **12** in the first, second, third, . . . , $(m-1)$ -th, and m -th rows in order.

In general, the scanning signal supplied to the scanning line **12** in the i -th row is represented as $Gwr(i)$.

[0043] The data signal output circuit **50** is a circuit for outputting a data signal to the pixel portions **110** located in the row selected by the scanning line driving circuit **120** via the data line **14** under the control of the control circuit **30**. The data signal is a voltage signal obtained by converting 10-bit video data $Vdata$ into analog data. That is, the data signal output circuit **50** converts video data $Vdata$ for one row corresponding to pixel portions **110** in the first to $(3n)$ -th columns in the selected row into analog data, and outputs the analog data to the first to $(3n)$ -th column of data lines **14** in this order.

[0044] Although not illustrated in the figure, a power supply circuit is provided outside the display region **100**, and generates potentials V_{el} and V_{ct} of a power supply for the control circuit **30**, the scanning line driving circuit **120**, the data signal output circuit **50**, and the OLED.

[0045] In the figure, the data signals output to the first, second, third, . . . , $(3n-2)$ -th, $(3n-1)$ -th, and $(3n)$ -th column of data lines **14** are denoted in order by $Vd(1)$, $Vd(2)$, $Vd(3)$, . . . , $Vd(3n-2)$, $Vd(3n-1)$, and $Vd(3n)$. Generally, for example, the potential of the data line **14** in the $(3j-2)$ -th column is denoted by $Vd(3j-2)$.

[0046] FIG. **3** is a diagram illustrating an electrical configuration of the pixel portion in the electro-optical device **10**.

[0047] The pixel portions **110R**, **110G**, and **110B** have the same configuration from an electrical perspective. Therefore, an electrical configuration of the pixel portions **110R**, **110G**, and **110B** will be described with the pixel portion **110R** corresponding to the i -th row and $(3j-2)$ -th column as an example.

[0048] As shown in the figure, the pixel portion **110R** includes P-channel MOS transistors **121** and **122**, an OLED **130**, and a capacitance element **140** from an electrical perspective.

[0049] In the description of the pixel portion, the term “from an electrical perspective” is used to refer to a plurality of elements constituting the pixel portion and a coupling relationship between the plurality of elements.

[0050] In the OLED **130** of the pixel portion **110R**, a light-emitting layer **132R** is sandwiched between a pixel electrode **131** and a common electrode **133**. The light-emitting layer **132R** emits light including an R wavelength range. The pixel electrode **131** functions as an anode, and the common electrode **133** functions as a cathode. In the OLED **130**, when a current flows from the anode to the cathode, holes injected from the anode and electrons injected from the cathode are recombined in the light-emitting layer **132R** to generate excitons, and light including the R wavelength range is generated.

[0051] In the OLED **130** of the pixel portion **110G**, the light-emitting layer **132G** is sandwiched between the pixel electrode **131** and the common electrode **133**. The light-emitting layer **132G** emits light including a G wavelength range. In the OLED **130** of the pixel portion **110B**, the light-emitting layer **132B** is sandwiched between the pixel electrode **131** and the common electrode **133**. The light-emitting layer **132B** emits light including a B wavelength range.

[0052] Each of the light-emitting layers **132R**, **132G**, and **132B** includes at least a light-emitting functional layer that emits light of each color. The light-emitting layers **132R**, **132G**, and **132B** may have a configuration in which one or more organic layers other than the light-emitting functional layer are sandwiched.

[0053] In the transistor **121** of the pixel portion **110R** in the i -th row and the $(3j-2)$ -th column, a gate node g is coupled to the drain node of the transistor **122**, a source node is coupled to the power supply line **116** for the potential V_{el} , and a drain node is coupled to the pixel electrode **131** which is the anode of the OLED **130**.

[0054] In the transistor **122** of the pixel portion **110R** in the i -th row and the $(3j-2)$ -th column, a gate node is coupled to the scanning line **12** in the i -th row, and a source node is coupled to the data line **14** of the $(3j-2)$ -th column. The common electrode **133** functioning as the cathode of the

OLED **130** is coupled to the power supply line **118** of a potential Vct. Further, since the electro-optical device **10** is formed at a silicon substrate, a substrate potential of the transistors **121** and **122** is set to, for example, a potential equivalent to the potential Vel.

[0055] The pixel portion **110R** illustrated in FIG. **3** is common to the pixel portions **110G** and **110B** from an electrical perspective. However, the light-emitting layer **132R** is replaced with the light-emitting layer **132G** in the pixel portion **110G**, and is replaced with the light-emitting layer **132B** in the pixel portion **110B**.

[0056] The X direction in FIGS. **1**, **2**, and **3** is a direction in which the scanning lines **12** extend in the electro-optical device **10**, and indicates a horizontal direction on a display screen. The Y direction is a direction in which the data lines **14** extend and indicates a vertical direction on the display screen. A two-dimensional plane determined by the X direction and the Y direction is a substrate surface of the semiconductor substrate. A Z direction in FIG. **1** is perpendicular to the X direction and the Y direction, and is a direction in which light is emitted from the OLED **130**. Further, in the present description, a plan view refers to viewing the semiconductor substrate from a direction opposite to the Z direction, and a cross-sectional view refers to viewing the semiconductor substrate cut in a direction perpendicular to the substrate surface.

[0057] FIG. **4** is a timing chart for describing an operation of the electro-optical device **10**.

[0058] In the electro-optical device **10**, the m rows of scanning lines **12** are scanned row by row in an order of first, second, third, . . . , m-th row in a period of a frame (V). Specifically, as shown in the figure, the scanning signals /Gwr(1), /Gwr(2), . . . , /Gwr(m-1), . . . , /Gwr(m) are sequentially and exclusively set to an L level by the scanning line driving circuit **120** for each horizontal scanning period (H).

[0059] In the present disclosure, periods in which adjacent scanning signals among scanning signals /Gwr(1) to /Gwr(m) become at an L level are separated in time. Specifically, after a scanning signal /Gwr(i-1) changes from an L level to a H level, the next scanning signal /Gwr(i) becomes at an L level after a period. This period corresponds to a horizontal blanking period.

[0060] In the present description, the period of the one frame (V) refers to a period required to display one frame of the image designated by the video data Vid. When a length of the period of the one frame (V) is the same as a vertical synchronization period, for example, when a frequency of the vertical synchronization signal included in the synchronization signal Sync is 60 Hz, the length is 16.7 milliseconds corresponding to one cycle of the vertical synchronization signal. Further, the horizontal scanning period (H) is a time interval in which the scanning signals /Gwr(1) to /Gwr(m) sequentially become at an L level, but for convenience in the figure, a start timing of the horizontal scanning period (H) is set as approximately a center of the horizontal blanking period.

[0061] When a certain scanning signal among the scanning signals /Gwr(1) to /Gwr(m), for example, the scanning signal /Gwr(i) supplied to the scanning line **12** in the i-th row becomes an L level, the transistor **122** enters an ON state in the pixel portion **110R** of the i-th row and the (3j-2)-th column. As a result, the gate node g of the transistor **121** in the pixel portion **110R** is electrically coupled to the data line **14** of the (3j-2)-th column.

[0062] In the present description, an “ON state” of the transistor means that the source node and drain node of the transistor are electrically closed and enters a low impedance state. Moreover, an “off state” of the transistor means that the source node and drain node are electrically open and enters a high impedance state.

[0063] Further, in the present description, “electrically coupled” or simply “coupled” means a state in which two or more elements are directly or indirectly coupled or connected. “Electrically decoupled” or simply “decoupled” means a state in which two or more elements are not directly or indirectly coupled or connected.

[0064] In the horizontal scanning period (H) in which the scanning signal /Gwr(i) becomes at the L level, the data signal output circuit **50** converts the video data Vdata decomposed into R, G, and B

into analog potentials $V_d(1)$ to $V_d(3n)$ and outputs the analog potentials $V_d(1)$ to $V_d(3n)$ as data signals to the data lines **14** in the first to $(3n)$ -th columns. The video data V_{data} decomposed into R, G, and B is three primary color components of the gradation levels of the pixels in the first to i -th rows and the first to n th columns indicated by the video data V_{id} .

[0065] For example, in the case of the $(3j-2)$ -th column, the data signal output circuit **50** converts the R gradation level $R(i, j)$ of the pixel in the i -th row and the j -th column indicated by the video data V_{id} into an analog signal potential $V_d(3j-2)$ and outputs the analog signal potential $V_d(3j-2)$ as a data signal to the data line **14** in the $(3j-2)$ -th column.

[0066] In the horizontal scanning period (H) when the scanning signal $/G_{wr}(i-1)$ one row before the scanning signal $/G_{wr}(i)$ becomes at the L level, the data signal output circuit **50** converts an R gradation level $R(i-1, j)$ of the pixel in a $(i-1)$ -th row and a j -th column into an analog signal potential $V_d(3j-2)$ and outputs the analog signal potential $V_d(3j-2)$ as a data signal to the $(3j-2)$ -th column data line **14**.

[0067] The data signal of the potential $V_d(3j-2)$ is applied to the gate node g of the transistor **121** in the pixel portion **110R** in the i -th row and the $(3j-2)$ -th column via the data line **14** in the $(3j-2)$ -th column, and the potential $V_d(3j-2)$ is held in the capacitance element **140**. Therefore, the transistor **121** causes a current according to a voltage between the gate node and the source node to flow through the OLED **130**.

[0068] Even when the scanning signal $G_{wr}(i)$ becomes at the H level and the transistor **122** is turned off, the potential $V_d(3j-2)$ is held in the capacitance element **140**, so that a current continues to flow through the OLED **130**. Therefore, in the pixel portion **110R** in the i -th row and $(3j-2)$ -th column, the OLED **130** continues to emit light at the voltage held in the capacitance element **140**, that is, at a brightness corresponding to the gradation level until the period of the one frame (V) has elapsed, the transistor **122** is turned on again, and a voltage of the data signal is applied again.

[0069] Although the pixel portion **110R** in the i -th row and $(3j-2)$ -th column has been described here, the OLEDs **130** of the pixel portions **110R**, **110G**, and **110B** in the i -th row and the columns other than the $(3j-2)$ -th column also emit light at a luminance indicated by the video data V_{data} .

[0070] Further, the OLEDs **130** of the pixel portions **110R**, **110G**, and **110B** in the rows other than the i -th row also emit light at the luminance indicated by the video data V_{data} as the scanning signals $/G_{wr}(1)$ to $/G_{wr}(m)$ sequentially becomes at the L level.

[0071] Therefore, in the electro-optical device **10**, the OLEDs **130** in all the pixel portions **110R**, **110G**, and **110B** from the first row and the first column to the m -th row and the $(3n)$ -th column emit light at the luminance indicated by the video data V_{data} in the period of the one frame (V), and one frame of an image is displayed.

[0072] FIG. 5 is a plan view illustrating an example of disposition of the pixel portions **110R**, **110G**, and **110B** in the electro-optical device **10**, and FIG. 6 is a cross-sectional view of main parts taken along line A-A' in FIG. 5.

[0073] As illustrated in FIG. 5 and as described above, the pixel portions **110R**, **110G**, and **110B** are arranged side by side in the X direction repeatedly in this order in plan view.

[0074] In FIG. 6, the substrate **102** is a semiconductor substrate such as silicon. The substrate **102** is provided with a circuit layer **143**. The circuit layer **143** is provided to correspond to the pixel portion **110R**, **110G**, or **110B**, and elements such as the transistors **121** and **122**, or various wirings are provided.

[0075] An insulating layer **103** is provided on the substrate **102**. Contact holes H2 are provided in the insulating layer **103**. The contact holes H2 are filled with a coupling member **147** such as tungsten.

[0076] A laminated body of a reflective electrode **171** and a pixel electrode **131** is provided for each of the pixel portions **110R**, **110G**, and **110B**.

[0077] Specifically, a metal wiring layer having light reflectivity such as Al, an alloy thereof, or Ag is formed at the insulating layer **103** filled with the coupling member **147**. After the film formation,

the metal wiring layer is in contact with the coupling member **147**, and the reflective electrode **171** is provided by patterning of a rectangular shape in plan view.

[0078] A transparent conductive layer having optical transparency and conductivity, such as indium tin oxide (ITO) is formed to cover the insulating layer **103** and the reflective electrode **171**. After the film formation, the pixel electrode **131** is provided by patterning such that the transparent conductive layer overlaps the reflective electrode **171** and, in plan view, is positioned inside the periphery of the reflective electrode **171**. The pixel electrode **131** is light transmittance.

[0079] When Al is used for the reflective electrode **171**, it is preferable to provide a conductive material such as TiN as a barrier layer with a thickness of about several nm between the reflective electrode **171** and the pixel electrode **131** made of ITO.

[0080] Since the reflective electrode **171** comes into contact with the coupling member **147**, the pixel electrode **131** is electrically coupled to the drain node of the transistor **121** included in the circuit layer **143** via the reflective electrode **171** and the coupling member **147**.

[0081] The pixel separation layer **104** having light transmittance and an insulative property is provided to cover the insulating layer **103**, the reflective electrode **171**, and the pixel electrode **131**. Then, an opening region Ar that exposes the pixel electrode **131** by patterning is provided in a pixel separation layer **104**. Specifically, in plan view, the opening region Ar has a rectangular shape defined by an opening end Ap, and in cross-sectional view, the opening region Ar is provided to overlap the periphery of the pixel electrode **131**, as illustrated in FIG. 6.

[0082] After patterning of the pixel separation layer **104**, a partition wall **161** and an upper portion **163** are provided.

[0083] FIG. 7 is a cross-sectional view illustrating a stage in the manufacturing process of the electro-optical device **10** where a partition wall **161** and an upper portion **163** are provided.

[0084] The partition wall **161** and the upper portion **163** are provided, for example, by integrated patterning. Specifically, the partition wall **161** and the upper portion **163** are provided at boundaries between the adjacent pixel portions **110R**, **110G**, and **110B**, as indicated by hatching in FIG. 5, in plan view.

[0085] The partition wall **161** and the upper portion **163** are provided in a lattice shape with a portion extending along the X direction and a portion extending along the Y direction in plan view. Therefore, the pixel portions **110R**, **110G**, and **110B** are surrounded by the partition wall **161** and the upper portion **163** in plan view.

[0086] The partition wall **161** is made of, for example, a conductive metal wiring layer such as aluminum. The upper portion **163** is made of a conductive metal wiring layer such as titanium, which is made of a material that has a lower etching rate than the partition wall, that is, is less susceptible to etching. The partition wall **161** has a light-shielding property against light emitted by the light-emitting layer **132**.

[0087] In the integrated etching, since the etching of the partition wall **161** progresses faster than that of the upper portion **163**, the upper portion **163** is wider than the partition wall **161** in plan view, and both ends of the upper portion **163** have a so-called overhang structure in which the ends protrude from a side of the partition wall **161** in cross-section view.

[0088] The side of the partition wall **161** has a tapered shape in cross-section view as will be described later, that is, a width narrowed in the X direction or Y direction toward an upper surface in the figure, but is not illustrated in the tapered shape in FIGS. 6 and 7 for simplified description. Further, the partition wall **161** extends to the outside of the display region **100** and is electrically coupled to the output terminal of the power supply circuit. Therefore, the partition wall **161** is maintained at the potential Vct generated by the power supply circuit.

[0089] Further, the upper portion **163** is made of a conductive metal wiring layer, but may be made of a material having an insulative property.

[0090] FIG. 8 is a cross-sectional view illustrating a manufacturing process of the electro-optical device **10** immediately after the formation of the light-emitting layer **132R**.

[0091] The light-emitting layer **132R** is deposited from above in the figure. Therefore, the light-emitting layer **132R** is formed to cover the opening region **Ar** of the pixel separation layer **104** in the pixel portion **110R**, and is also formed at an upper surface of the upper portion **163**.

[0092] In other words, the light-emitting layer **132R** is formed in the pixel portion **110R** to overlap the pixel electrode **131**, using the upper portion **163**, which has been already provided, as a mask. Therefore, since the light-emitting layer **132R** is formed by self-alignment, not by photolithography, an exposure process using an expensive fine metal mask is not required.

[0093] The light-emitting layer **132R** is also provided in the pixel portions **110G** and **110B** for different colors at this stage, but is removed by subsequent etching.

[0094] After the light-emitting layer **132R** is formed, the common electrode **133** is provided by forming a conductive layer having transparency, reflectivity, and conductivity. The common electrode **133** comes into contact with a side wall of the partition wall **161**. Therefore, the common electrode **133** is maintained at the potential V_{ct} via the partition wall **161**.

[0095] In this stage, the light-emitting layer **132R**, the common electrode **133**, and a sealing layer **155** are provided in an overlapped manner on the pixel electrode **131** in the pixel portions **110G** and **110B**.

[0096] In the pixel portion **110G**, the pixel portion **110R** is first covered with and protected by photoresist in order to obtain the light-emitting layer **132G** for a corresponding correct color. Then, the light-emitting layer **132R**, the common electrode **133**, and the sealing layer **155** in the pixel portions **110G** and **110B** are removed by etching to expose the pixel electrode **131**.

[0097] The G light-emitting layer **132G** is formed by self-alignment using the upper portion **163** as a mask, and the common electrode **133** and sealing layer **155** are provided in an overlapped manner, similar to the pixel portion **110R**, in the pixel portion **110G**.

[0098] In this stage, in the pixel portion **110B**, the light-emitting layer **132G**, the common electrode **133**, and the sealing layer **155** are provided in an overlapped manner on the pixel electrode **131**.

[0099] In the pixel portion **110B**, the pixel portion **110G** is first covered with and protected by photoresist to obtain the light-emitting layer **132B** for a corresponding correct color. Then, the light-emitting layer **132G**, the common electrode **133**, and the sealing layer **155** in the pixel portion **110B** are removed by etching to expose the pixel electrode **131**.

[0100] In the pixel portion **110B**, the B light-emitting layer **132B** is formed by self-alignment using the upper portion **163** as a mask, and the common electrode **133** and the sealing layer **155** are provided in an overlapped manner, similar to the pixel portions **110R** and **110G**.

[0101] Accordingly, the configuration as illustrated in FIG. **6** is obtained.

[0102] In FIG. **6**, the light-emitting layers **132R**, **132G**, and **132B** and the same conductive layer as the common electrode **133** are overlapped on the upper surface of the upper portion **163**, but are spaced near the boundaries of the pixel portions **110R**, **110G**, and **110B**. This space is caused by etching for protection of the photoresist.

[0103] Then, one or more sealing layers and planarizing layers having an insulative property are provided.

[0104] In the electro-optical device **10**, even when the light generated in the light-emitting layers **132R**, **132G**, and **132B** travels in the direction opposite to the Z direction, the light is reflected by the reflective electrode **171** and emitted in the Z direction.

[0105] The present embodiment adopts a so-called overhang structure in which the upper portion **163** protrudes from the partition wall **161**. This overhang structure makes it possible to form the light-emitting layers **132R**, **132G**, and **132B** in a self-aligned manner.

[0106] However, on the other hand, when the overhang structure is used, there are problems such as blocking of emitted light and a decrease in aperture ratio.

[0107] Since no color distinguishment will be described hereinafter, the pixel portion is denoted by reference number **110** and the light-emitting layer is denoted by reference number **132**.

[0108] FIG. **9** is an enlarged cross-sectional view illustrating the pixel portion **110** in the electro-

optical device **10**, and illustrates up to a stage in which the common electrode **133** has been formed. The angle θ is formed between a line F1 that couples the opening end Ap to a tip Bp of the upper surface of the upper portion **163** that protrudes from the partition wall **161** and a line F2 passing through the opening end Ap and directed to a substrate normal direction (Z direction).

[0109] A region of the light-emitting layer **132** that emits light when a voltage is applied is a region where the light-emitting layer **132R** is sandwiched between the pixel electrode **131** and the common electrode **133**, that is, the opening region Ar where the light-emitting layer **132** is in contact with the pixel electrode **131**.

[0110] Disposition intervals in the X direction of the partition wall **161** is equal to a pitch of the pixel portions **110R**, **110G**, and **110B**. In a micro-display in which the pitch of the pixel portion is about 2 to 3 μm , when a height of the partition wall **161** and the upper portion **163** is about 0.5 to 0.7 μm , when the angle θ becomes large, a width W of the opening region Ar becomes extremely narrow, and the aperture ratio decreases. When the aperture ratio decreases, it is necessary to increase the luminance of the light-emitting layer **132** to ensure brightness. In order to increase the luminance, a current density when the light-emitting layer **132** is driven may be increased.

However, when the light-emitting layer **132** is driven with a high current density, the deterioration of the light-emitting layer **132** progresses exponentially. Specifically, when the current density increases four times, a luminance life becomes 1/10 or less. When the luminance life is shortened, not only a screen will become darker, but also the shorter luminance life will have an adverse effect on the quality of products, such as burn-in of the display. The luminance life is a period until the luminance decreases by half when the light-emitting layer **132** is driven under the same conditions.

[0111] FIG. **10** is a diagram illustrating an example of luminance change with respect to an observation angle in the electro-optical device.

[0112] In the figure, the observation angle on a horizontal axis is an angle with respect to the substrate normal direction when the Z direction of the substrate normal direction is set to 0 degrees. Further, in the figure, relative luminance on a vertical axis is a value of relative luminance obtained by normalizing luminance at which the observation angle is 0 degrees by 1.0.

[0113] As shown in the figure, the relative luminance decreases as the observation angle increases. An angle at which the luminance is 50% of the luminance at which the observation angle is 0 degrees is in a range of 30 to 40 degrees, and the observation angle at which the luminance is 25% of the luminance at 0 degrees is about 45 degrees. When the observation angle is 45 degrees or more, not only the relative luminance will become low and the display become unsuitable for use, but the decrease in the aperture ratio becomes pronounced as described above.

[0114] Therefore, in the micro-display, a limit of the angle θ at which the luminance can be ensured and the decrease in aperture ratio is curbed is 45 degrees.

[0115] On the other hand, as the angle θ becomes smaller, the aperture ratio increases. However, when the angle θ is too small, the light emitted from the light-emitting layer **132**, particularly the light emitted from the vicinity of the opening end Ap, is shielded by a portion Shd of the upper portion **163** that protrudes from the partition wall **161** as illustrated in FIG. **11**, and the use efficiency of the emitted light is degraded. The light shielded by the portion Shd is reflected by the portion Shd, and is subjected to multiple reflections by the side of the partition wall **161**, other members having reflectivity, an interface of the insulating layer, or the like, which causes stray light.

[0116] About 60% of the total amount of light emitted from the light-emitting layer **132** is emitted in a range of 0 to 20 degrees. Therefore, when the light emitted at an angle θ of 20 degrees is not shielded by the upper portion **163**, it is considered that the light emitted from the light-emitting layer **132** can be efficiently extracted to the outside of the electro-optical device **10**.

[0117] When the angle θ is smaller than 20 degrees, the luminance increases, but the high-luminance light is shielded by the portion Shd to cause multiple reflections, which is not preferable.

[0118] Therefore, when the angle θ is less than 45 degrees, it is possible to curb the decrease in

the aperture ratio while securing the required luminance. Further, when the angle θ is 20 degrees or more and 40 degrees or less, it is possible to increase the aperture ratio while securing a higher luminance.

[0119] For convenience, a distance of a component in the substrate normal direction (Z direction) of the straight line F1 coupling the opening end Ap to the tip Bp is defined as α , that is, a height of the tip Bp with reference to the opening end Ap is defined as α . Also, a distance of s component of the straight line F1 in a direction along the substrate surface is β , that is, a distance from the opening end Ap to the tip Bp when viewed in plan view is β .

[0120] The angle θ is expressed as $\tan.\sup.-1 (\beta/\alpha)$.

[0121] In other words, the angle θ being less than 45 degrees can be expressed as

[0122] $\alpha > \beta$. Also, the angle θ being 20 degrees or more and 40 degrees or less can be expressed as

$\tan 20^\circ \leq (\beta/\alpha) \leq \tan 40^\circ$.

[0123] The common electrode **133** is an example of a “first electrode”, and the pixel electrode **131**, or a laminated body of the reflective electrode **171** and the pixel electrode **131**, is an example of a “second electrode”.

[0124] FIG. **12** is a cross-sectional view illustrating main parts of the electro-optical device **10** according to the first embodiment. In the first embodiment, the angle θ is set to 30 degrees.

[0125] In this example, since the angle θ is 30 degrees, when α is 0.6 μm , β is about 0.346 μm . When the angle θ is 30 degrees, the width W of the opening region Ar is about 1.5 μm .

[0126] About 80% of a total amount of light emitted from the light-emitting layer **132** is emitted in a range of 0 degrees to 30 degrees, but light between 20 degrees and 30 degrees is emitted without being blocked by the portion Shd. Therefore, in the first embodiment, the light emitted from the light-emitting layer **132** can be used efficiently.

[0127] FIG. **13** is a cross-sectional view illustrating main parts of the electro-optical device **10** according to a second embodiment. In the second embodiment, the angle θ is set to 20 degrees. In the second embodiment, since the aperture ratio is higher than in the first embodiment, it is possible to secure high luminance. Therefore, the second embodiment is advantageous in terms of the luminance life of the light-emitting layer compared to the first embodiment.

[0128] FIG. **14** is a cross-sectional view illustrating main parts of an electro-optical device **10** according to a third embodiment. In the third embodiment, the angle θ is set to 40 degrees. In the third embodiment, the aperture ratio is smaller than in the first embodiment, but the amount of light shielded by portion Shd is reduced, and as a result, a more amount of light can be emitted from the light-emitting layer **132**. Specifically, in the third embodiment, it is possible to emit about 90% of the light emitted from the light-emitting layer **132**.

[0129] FIG. **15** is a cross-sectional view illustrating main parts of an electro-optical device according to a comparative example for comparison with the first to third embodiments. In this comparative example, the angle θ is set to 50 degrees, which is larger than 45 degrees. When the angle θ is 50 degrees, about 95% of the light emitted from the light-emitting layer **132** can be emitted, but the width W of the opening region Ar is about $\frac{1}{2}$ and an area is about $\frac{1}{4}$ compared to that in the first embodiment. Therefore, in the comparative example, it is necessary for a current density of the light-emitting layer **132** to be about four times that of the first embodiment in order to ensure the same luminance as in the first embodiment. As described above, since the luminance life becomes 1/10 when the current density increases four times, the luminance life becomes shorter, and when the luminance life becomes shorter, not only the screen will become dark due to aging, but also the shorter luminance life will have an adverse effect on the quality of products, such as burn-in of the display.

[0130] In the first to third embodiments described above (hereinafter referred to as an “embodiment or the like”), various modifications or applications are possible as follows.

[0131] FIG. **16** is a partial cross-sectional view illustrating main parts of the electro-optical device

10 according to a first application example of the embodiment or the like.

[0132] In the first application example, pixel portions **110R**, **110G**, and **100B** are provided with an optical resonance structure corresponding to the color. The optical resonance structure refers to a structure in which, when an optical distance between a reflective surface of the common electrode **133** and a reflective surface of the reflective electrode is L_r in the pixel portion **110R**, L_g in the pixel portion **110G**, and L_b in the pixel portion **110B**, the optical distances L_r , L_g , and L_b are set to a distance associated with the wavelengths of each color. Further, a specific example of setting the optical distances L_r , L_g , and L_b to a distance associated with the wavelengths of each color is considered to be a next measure. That is, [0133] a first measure is to make thicknesses of the light-emitting layers **132R**, **132G**, and **132B** different for each color.

[0134] A second measure is to make a thickness of the pixel electrode **131** having light transmittance different for each color.

[0135] A third measure is to make a sum of the thickness of the pixel electrode **131** and the thickness of the light-emitting layers **132R**, **132G**, or **132B** different for each color. FIG. **16** illustrates the first measure among the measures.

[0136] The optical distances L_r , L_g , and L_b have a relationship $L_r > L_g > L_b$.

[0137] In the optical resonance structure, light emitted from the light-emitting layer **132R**, **132G**, or **132B** resonates due to reflection between the reflective electrode **171** and the common electrode **133**, and is emitted at a resonance wavelength that has been set to correspond to the color R, G, or B.

[0138] Therefore, in the first application example with the optical resonance structure, light having a wavelength corresponding to a color is enhanced and emitted, so that a spectrum is sharpened and intensified, enabling an improvement in both color purity and brightness.

[0139] The light-emitting layer **132R**, **132G**, or **132B** has a laminated structure of a hole injection layer, a hole transport layer, a light-emitting functional layer, an electron blocking layer, an electron transport layer, and an electron injection layer, which are not actually shown in the figure.

Therefore, it is possible to make a film thickness of the light-emitting layer **132R**, **132G**, or **132B** different for each color by adjusting the thicknesses of the layers for each color.

[0140] Strictly speaking, the optical distance is a value obtained by multiplying a distance between the reflective electrode **171** and the common electrode **133** by a refractive index of the pixel electrode **131** and the light-emitting layer, which are media between the reflective electrode **171** and the common electrode **133**, but is simply shown as a physical distance in the figure.

[0141] Further, in the first application example, the pixel electrode **131** having transparency can be omitted. When the pixel electrode **131** is omitted, the optical distances L_r , L_g , and L_b may be set to distances associated with the wavelengths of the respective colors by using the first measure.

[0142] FIG. **17** is a partial cross-sectional view illustrating main parts of an electro-optical device **10** according to a second application example.

[0143] The second application example is the same as the first application example in that the pixel portions **110R**, **110G**, and **100B** have optical resonance structures corresponding to the colors. However, the second application example differs from the first application example in that the optical distances L_r , L_g , and L_b are adjusted by a film thickness of an insulating layer provided between the pixel electrode **131** and the reflective electrode **171**.

[0144] FIG. **17** illustrates an example in which the insulating layer between the pixel electrode **131** and the reflective electrode **171** is not provided in the pixel portion **110B**, is one layer in the pixel portion **110G**, and includes two layers obtained by adding another insulating layer to the insulating layer, that is the one layer provided in the pixel portion **110G**, in the pixel portion **110R**.

[0145] Since an electric field becomes weaker when the thickness of the light-emitting layer **132** increases, a higher voltage needs to be applied to obtain the same luminance, but in the second application example, the optical distance can be set to $L_r > L_g > L_b$ in a state where the thicknesses of the light-emitting layers **132R**, **132G**, and **132G** are uniform. Therefore, in the second application

example, it is not necessary to drive the R light-emitting layer **132R** having the longest optical distance with a high voltage in order to increase the color purity and the luminance.

[0146] In the embodiment, an opening shape of the opening region Ar in the pixel portions **110R**, **110G**, and **110B** is a rectangular shape, but the present disclosure is not limited thereto. For example, the opening shape may be a polygon such as a hexagon or may be a circle, an ellipse, or the like. Further, an opening area of the opening region Ar may not be uniform in the pixel portions **110R**, **110G**, and **110B**, but may be different for each color. For example, $B > G > R$ for the opening area of the opening region Ar.

[0147] The pixel portions **110R**, **110G**, and **110B** may be aligned in the X direction or may be aligned in the Y direction. Further, the pixel portions **110R** and **110B** may be arranged in the same columns, and the pixel portion **110G** may be arranged in a column adjacent to the column of the pixel portions **110R** and **110B**.

[0148] In the description of the embodiment or the like, the light-emitting layers **132R**, **132G**, and **132B** are formed in this order, but the order of layer formation is not limited thereto.

[0149] Next, an electronic apparatus to which the electro-optical device **10** according to the embodiment is applied will be described. The electro-optical device **10** is suitable for application of a small pixel size and high definition display. Consequently, a head-mounted display will be described as an example of the electronic apparatus.

[0150] FIG. **18** is a diagram illustrating an appearance of a head-mounted display, and FIG. **19** is a diagram illustrating an optical configuration thereof.

[0151] First, as illustrated in FIG. **18**, the head-mounted display **300** includes temples **310**, a bridge **320**, and lenses **301L** and **301R** in appearance, similar to ordinary glasses. Further, the head-mounted display **300** includes an electro-optical device **10L** for a left eye and an electro-optical device **10R** for a right eye near the bridge **320** and behind the lenses **301L** and **301R** (below in the figure), as illustrated in FIG. **19**.

[0152] An image display surface of the electro-optical device **10L** is disposed to the left in FIG. **19**. Accordingly, an image displayed by the electro-optical device **10L** is output in a 9 o'clock direction in the figure via the optical lens **302L**. A half mirror **303L** reflects an image displayed by the electro-optical device **10L** in the 6 o'clock direction, and transmits light incident from a 12 o'clock direction. An image display surface of the electro-optical device **10R** is disposed to the right opposite to the electro-optical device **10L**. Accordingly, an image displayed by the electro-optical device **10R** is emitted in a 3 o'clock direction in the figure through an optical lens **302R**. A half mirror **303R** reflects an image displayed by the electro-optical device **10R** in the 6 o'clock direction, and transmits light incident from the 12 o'clock direction.

[0153] In this configuration, a wearer of the head-mounted display **300** can observe the display images of the electro-optical devices **10L** and **10R** in a see-through state in which the display images are superimposed on an outside world.

[0154] Furthermore, in the head-mounted display **300**, an image for the left eye is displayed by the electro-optical device **10L**, and an image for the right eye is displayed by the electro-optical device **10R** in the images for both eyes involving parallax, so that it enables the wearer to sense the displayed image as having depth or stereoscopic effect.

[0155] The electronic apparatus including the electro-optical device **10** can be applied to an electronic viewfinder in a video camera or an interchangeable lens digital camera, a smart watch, a display unit of a wearable device, a light bulb in a projection projector, and the like, in addition to the head-mounted display **300**.

[0156] From the aspects illustrated above, the following aspects can be ascertained, for example. Hereinafter, for ease of understanding of each aspect, the reference signs in the drawings are conveniently written in parentheses, but this is not intended to limit the aspects to those shown.

[0157] An electro-optical device (**10**) according to one embodiment 1 includes a substrate (**102**), a first electrode (**133**), a second electrode (**131**, **171**) provided between the substrate (**102**) and the

first electrode (133), a pixel separation layer (104) having an insulative property and covering the periphery of the second electrode (131, 171), and opening in an opening region (Ar) overlapping the second electrode (131, 171) in plan view, a light-emitting layer (132) provided between the first electrode (133) and the second electrode (131, 171) and being in contact with the second electrode (131, 171) in the opening region (Ar), a partition wall (161) having a light-shielding property against light emitted by the light-emitting layer (132) and surrounding the first electrode (133), the light-emitting layer (132), and the second electrode (131, 171) in plan view, and an upper portion (163) provided so as to protrude from the partition wall in cross-sectional view on the upper surface of the partition wall (161) and having a light-shielding property against light emitted by the light-emitting layer (132), wherein, when a distance of a normal component to the substrate (102) of a shortest straight line coupling a tip of the upper portion (163) to an opening end (Ap) of the opening region (Ar) in cross-sectional view is defined as x , and a distance of a component of the straight line along a surface of the substrate (102) is β , there is a relationship of $\alpha > \beta$.

[0158] According to the electro-optical device of aspect 1, when $\alpha > \beta$, that is, the angle θ is greater than 45 degrees, it is possible to ensure the luminance and to curb the decrease in the aperture ratio due to the narrowing of the opening region.

[0159] In the electro-optical device (10) according to specific aspect 2 of aspect 1, a and B have a relationship of $\tan 20^\circ \leq (\beta/\alpha) \leq \tan 40^\circ$.

[0160] According to aspect 2, it is possible to curb the decrease in aperture ratio while ensuring high luminance, and to increase the use efficiency of the emitted light.

[0161] In the electro-optical device (10) according to another specific aspect 3 of aspect 1, the second electrode (131, 171) is a laminated body of a reflective electrode (171) and a pixel electrode (131) having reflectivity when viewed from the substrate (102), and the first electrode (133) has light transmittance and reflectivity.

[0162] In the electro-optical device (10) according to another specific aspect 4 of aspect 1, the second electrode (171) is a reflective electrode (171) having reflectivity, and the first electrode (133) has light transmittance and reflectivity.

[0163] The electro-optical device (10) according to another specific alternative aspect 5 of aspect 1 includes a sealing layer (155) having an insulative property and light transmittance and covering the first electrode (133).

[0164] In the electro-optical device (10) according to a specific alternative aspect 6 of aspect 1, the partition wall (161) has conductivity, and the first electrode (133) is in contact with the partition wall (161).

[0165] An electronic apparatus (300) according to aspect 8 includes the electro-optical device (10) according to any one of aspects 1 to 6.

Claims

1. An electro-optical device comprising: a substrate; a first electrode; a second electrode provided between the substrate and the first electrode; a pixel separation layer having an insulative property, covering a periphery of the second electrode, and opening in an opening region overlapping the second electrode in plan view; a light-emitting layer provided between the first electrode and the second electrode and being in contact with the second electrode in the opening region; a partition wall having a light-shielding property against light emitted by the light-emitting layer and surrounding the first electrode, the light-emitting layer, and the second electrode in plan view; and an upper portion provided so as to protrude from the partition wall in cross-sectional view on an upper surface of the partition wall, and having a light-shielding property against light emitted by the light-emitting layer, wherein when a distance of a normal component to the substrate of a shortest straight line coupling a tip of the upper portion to an opening end of the opening region in cross-sectional view is defined as a , and a distance of a component of the straight line along a

surface of the substrate is defined as β , there is a relationship of $\alpha > \beta$.

2. The electro-optical device according to claim 1, wherein α and β have a relationship of $\tan 20^\circ \leq (\beta/\alpha) \leq \tan 40^\circ$.

3. The electro-optical device according to claim 1, wherein the second electrode is a laminated body of a reflective electrode and a pixel electrode having reflectivity when viewed from the substrate, and the first electrode has light transmittance and reflectivity.

4. The electro-optical device according to claim 1, wherein the second electrode is a reflective electrode having reflectivity, and the first electrode has light transmittance and reflectivity.

5. The electro-optical device according to claim 1, comprising a sealing layer having an insulative property and light transmittance and covering the first electrode.

6. The electro-optical device according to claim 1, wherein the partition wall has conductivity, and the first electrode is in contact with the partition wall.

7. An electronic apparatus comprising the electro-optical device according to claim 1.
